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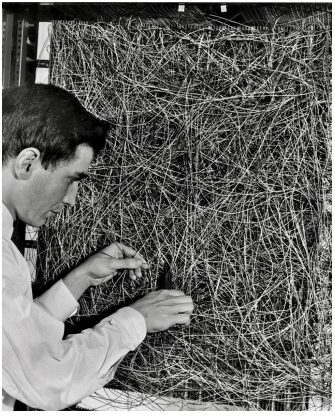
The Machine That Could Learn

Week 8 — Weighted Sums

Julian Garcia

FIT1061 — Introduction to Artificial Intelligence

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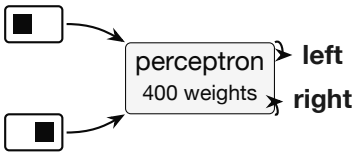


Frank Rosenblatt & the Mark I Perceptron.

MILESTONE

Rosenblatt's Perceptron

1958 · Cornell Aeronautical Laboratory

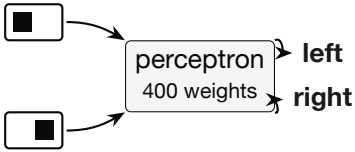


learns from ~**50** labelled cards — then labels cards it has never seen

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learns from ~**50** labelled cards — then labels cards it has never seen

> *"The Navy revealed the embryo of an electronic computer today that it expects will be able to walk, talk, see, write, reproduce itself, and be conscious of its existence."*
>
> — *The New York Times*, 1958

Can a machine learn to
distinguish one thing from another
— just from examples?

A neuron as maths

The weighted sum

Two classifiers, one dataset

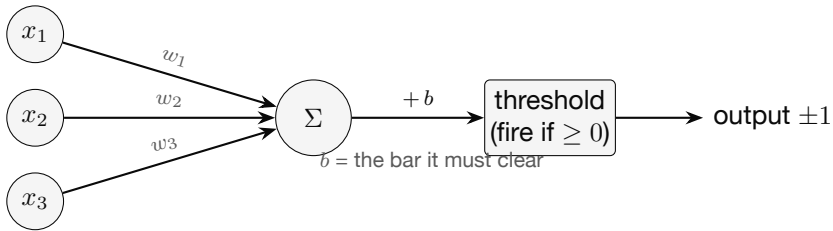
The pattern it could not learn

One machine, two limits

A neuron as maths

before the machine, an idea — a brain cell as arithmetic

The artificial neuron



The Mark I turned a card into $20 \times 20 = 400$ inputs.

The promise, and the exaggeration

What it actually did

Learn to tell a **left** square from a **right** one — after about 50 example cards.

The promise, and the exaggeration

What it actually did

Learn to tell a **left** square from a **right** one — after about 50 example cards.

What was claimed

“Capable of original thought… the first serious rival to the human brain ever devised.”

— *The New Yorker*, 1958

Today’s plan

1. The **weighted sum** — how one neuron decides.
2. **Two classifiers** on one dataset — same data, opposite mistakes.
3. The pattern it **could not learn**.
4. **Who** its decisions affect most.

A neuron as maths

The weighted sum

Two classifiers, one dataset

The pattern it could not learn

One machine, two limits

The weighted sum

how one neuron turns numbers into a decision

Last week's machine, on a small inbox

score$(c) = P(c) \times \prod_i P(w_i \mid c)$ — predict the **bigger score**; ties go to ham

A new inbox. The rule and the smoothing are last week's.

message	label	tracked words present
1	**spam**	prize
2	**spam**	prize, click, meeting
3	**spam**	prize, click, lunch
4	**spam**	click, winner
5	ham	prize, meeting, lunch
6	ham	click, meeting, lunch
7	ham	meeting
8	ham	lunch

Each row is one training message. The filter tracks five words; only those are shown.

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Each row is one training message. The filter tracks five words; only those are shown.

word w	$P(w \mid \text{spam})$	$P(w \mid \text{ham})$
prize	$4/6$	$2/6$
click	$4/6$	$2/6$
winner	$2/6$	$1/6$
meeting	$2/6$	$4/6$
lunch	$2/6$	$4/6$

Likelihoods with Laplace smoothing: $+1$ on top, $+2$ below. Four messages per class, so every denominator is 6.

Divide the two scores

The rule only asks which score is bigger. Divide one by the other:

$$\text{predict spam when } \frac{\text{score}(\text{spam})}{\text{score}(\text{ham})} > 1$$

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Both scores are products, so the ratio is a product of ratios. Message 2:

$$\frac{\text{score}(\text{spam})}{\text{score}(\text{ham})} = \underbrace{\frac{P(\text{spam})}{P(\text{ham})}}_{\text{prior (class sizes)}} \times \underbrace{\frac{P(\text{prize} \mid \text{spam})}{P(\text{prize} \mid \text{ham})}}_{\text{prize}} \times \underbrace{\frac{P(\text{click} \mid \text{spam})}{P(\text{click} \mid \text{ham})}}_{\text{click}} \times \underbrace{\frac{P(\text{meeting} \mid \text{spam})}{P(\text{meeting} \mid \text{ham})}}_{\text{meeting}}$$

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- A word's ratio says how much more often it appears in spam than in ham.
- Above 1 leans toward spam, below 1 toward ham. Multiply the ratios; above 1 overall means spam.
- Equal class sizes here, so the prior factor is 1.

One number per word

ratio above 1: the word **leans spam**; below 1: it **leans ham**

word w	$P(w \mid \text{spam})$	$P(w \mid \text{ham})$	ratio	the word leans
prize	$4/6$	$2/6$	2	**spam**
click	$4/6$	$2/6$	2	**spam**
winner	$2/6$	$1/6$	2	**spam**
meeting	$2/6$	$4/6$	$1/2$	ham
lunch	$2/6$	$4/6$	$1/2$	ham

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lunch	$2/6$	$4/6$	$1/2$	ham

message 2 (prize, click, meeting): $2 \times 2 \times \frac{1}{2} = 2 > 1 \;\rightarrow$ **spam**

the full scores, $2/27$ against $1/27$, give the same verdict

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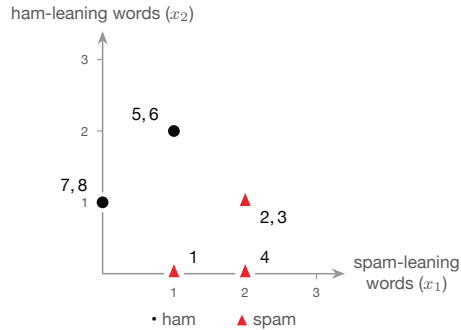
the full scores, $2/27$ against $1/27$, give the same verdict

winner and prize: different probabilities, same ratio, same lean.

Only the ratio reaches the verdict: **one number per word**.

One point per message

message	spam-leaning words (x_1)	ham-leaning words (x_2)	label
1	1	0	**spam**
2, 3	2	1	**spam**
4	2	0	**spam**
5, 6	1	2	ham
7, 8	0	1	ham

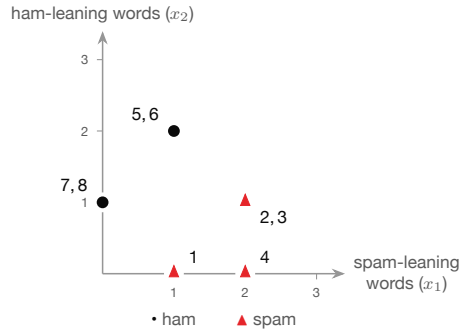


Count each message's words of each kind.

Messages 2 and 3 land on the same point.

One point per message

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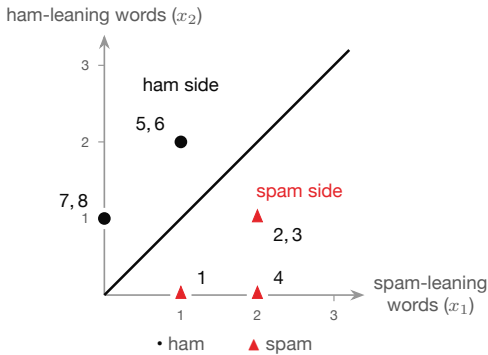
Count each message's words of each kind.

Messages 2 and 3 land on the same point.

All eight training messages are classified correctly.

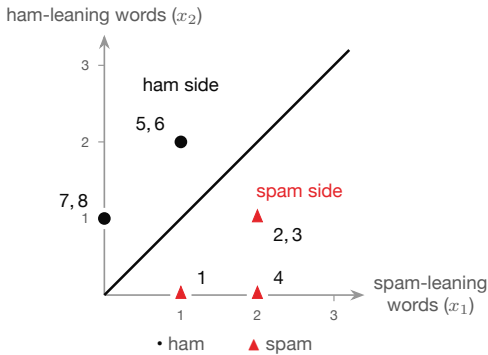
Where on this picture does the verdict flip?

Where the verdict flips



The verdict flips on the line $x_2 = x_1$: as many ham-leaning words as spam-leaning words.

Where the verdict flips



The verdict flips on the line $x_2 = x_1$: as many ham-leaning words as spam-leaning words.

Why this line?

A log turns the product into a sum

$\log(ab) = \log a + \log b$, so the product of ratios becomes a sum, one term per word:

$$\log_2(\text{score ratio}) = \log_2(\text{ratio}_{\text{prize}}) + \log_2(\text{ratio}_{\text{click}}) + \cdots$$

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$\log_2$ counts doublings: $\log_2 2 = +1$ $\log_2 1 = 0$ $\log_2 \frac{1}{2} = -1$

word w	ratio	$\log_2$ ratio	the word leans
prize	2	$+1$	**spam**
click	2	$+1$	**spam**
winner	2	$+1$	**spam**
meeting	$1/2$	-1	ham
lunch	$1/2$	-1	ham

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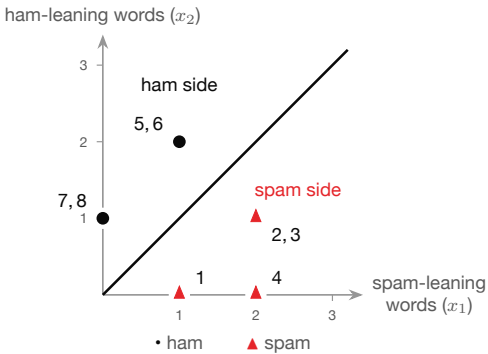
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message 2 (prize, click, meeting): $(+1) + (+1) + (-1) = +1 > 0 \rightarrow$ **spam**

"ratio above 1" has become "sum above 0".

Why the diagonal



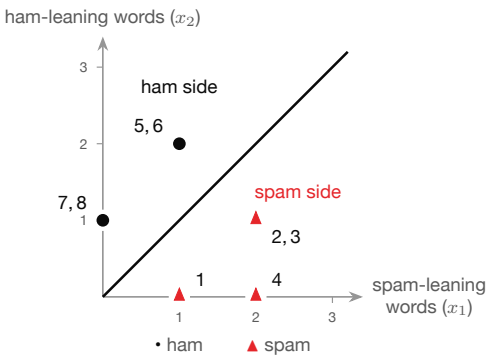
$+1$ per spam-leaning word, -1 per ham-leaning word:

$$\log_2(\text{score ratio}) = (+1) \cdot x_1 + (-1) \cdot x_2$$

Add up; the sign decides.

The sum is 0 exactly on $x_2 = x_1$: the diagonal.

Why the diagonal



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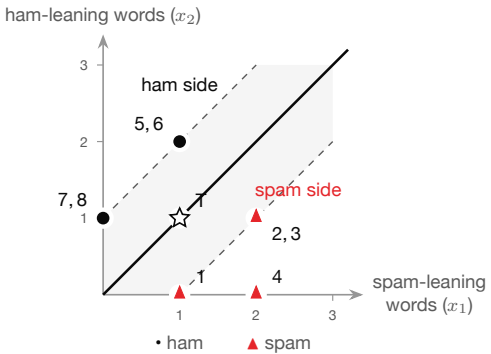
The sum is 0 exactly on $x_2 = x_1$: the diagonal.

a weighted sum, with weights set by counting

Is this the only line that works?

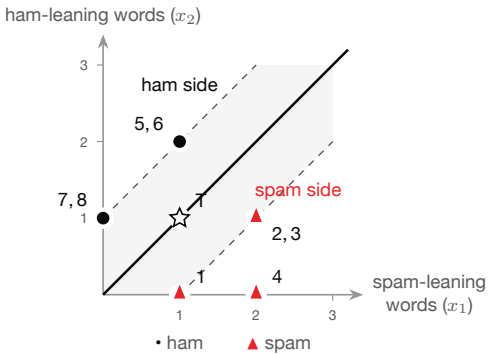
General rule: weight $= \log_2(\text{ratio})$. Two axes here only because every ratio is 2 or $1/2$.

Many lines fit the same eight messages



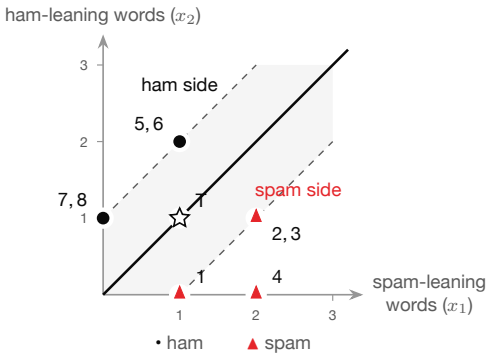
- Every line strictly inside the shaded band gets all eight right.
- Naive Bayes drew the solid line, through the origin: the prior factor is 1.

Many lines fit the same eight messages



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- New message T $=$ “prize, meeting” lands at $(1, 1)$, on the line: $+1 - 1 = 0$. Last week’s tie rule sends it to ham. **It is spam.**

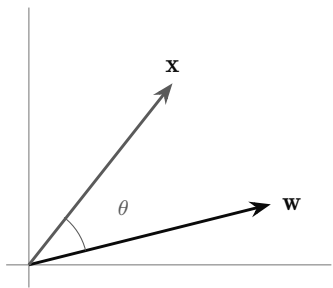
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This week the weights are set **by hand**, and the line moves.

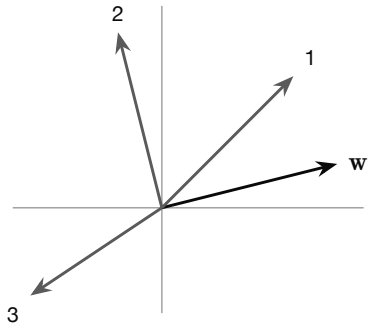
The dot product is the decision



- Arrows **agree** (small angle) → dot product **positive**.
- **Perpendicular** → dot product **zero**.
- Arrows **oppose** → dot product **negative**.

$$\mathbf{w} \cdot \mathbf{x} = w_1 x_1 + w_2 x_2$$

Where the sign comes from



	$\mathbf{x}$	$4x_1 + 1x_2$	
1	$(3, 3)$	$12 + 3 = +\mathbf{15}$	agree
2	$(-1, 4)$	$-4 + 4 = \mathbf{0}$	perpendicular
3	$(-3, -2)$	$-12 - 2 = \mathbf{-14}$	oppose

Same $\mathbf{w} = (4, 1)$ in all three rows — only $\mathbf{x}$ moves.

The perceptron decision rule

Weighted sum + threshold

$$\hat{y} = \begin{cases} +1 & \text{if } \mathbf{w} \cdot \mathbf{x} + b \geq 0 \\ -1 & \text{otherwise.} \end{cases}$$

$\hat{y}$ is the predicted class; y is the label the data came with.

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$\hat{y}$ is the predicted class; y is the label the data came with.

the warm-up's line: $\mathbf{w} = [1, -1]$, $b = 0$ gives score $= x_1 - x_2$.

e.g. $\mathbf{w} = [2, -1]$, $b = -1$ gives score $= 2x_1 - x_2 - 1$.

$\mathbf{w}$ and b are the model's **parameters** — the numbers that fix the decision.

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Note $\geq$, not $>$: a point exactly **on the boundary** fires $+1$.

Last week ties went to ham. Both are conventions, not evidence.

The forward pass — three operations

forward pass = pass the inputs through the neuron once to get a prediction

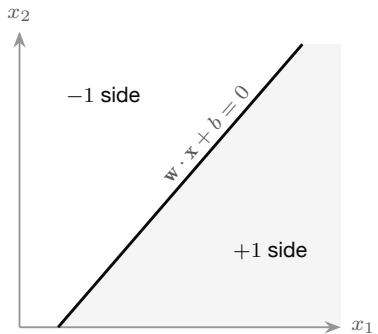
weighted_sum · predict · classify_all

```
function WEIGHTED_SUM(x, w, b):
    total ← b
    for each feature i:
        total ← total + w[i] * x[i]
    return total

function PREDICT(x, w, b):
    if WEIGHTED_SUM(x, w, b) ≥ 0:
        return +1
    else:
        return −1

function CLASSIFY_ALL(points, w, b):
    return [ PREDICT(x, w, b) for each x in points ]
```

The boundary is a line

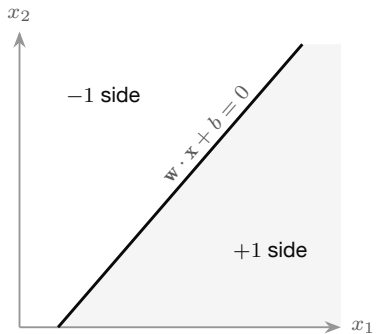


here w_1, w_2 have opposite signs, so $-w_1/w_2 > 0$.

The rule flips sign exactly where

$$\mathbf{w} \cdot \mathbf{x} + b = 0.$$

The boundary is a line



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Rearranged as a line:

$$x_2 = -\frac{w_1}{w_2}\, x_1 \; - \; \frac{b}{w_2}.$$

The slope is $-w_1/w_2$ — note the **minus** sign.

Worked example — the setup

$$\mathbf{w} = [\,2,\,-1\,], \quad b = -1$$

$$\text{score} = 2x_1 - x_2 - 1$$

the weighted sum $\mathbf{w} \cdot \mathbf{x} + b$ is the “score”

pt	(x_1, x_2)	label
A	$(2, 1)$	$+1$
B	$(3, 2)$	$+1$
C	$(3, 4)$	$+1$
D	$(1, 3)$	-1
E	$(0, 2)$	-1
F	$(0, 1)$	-1

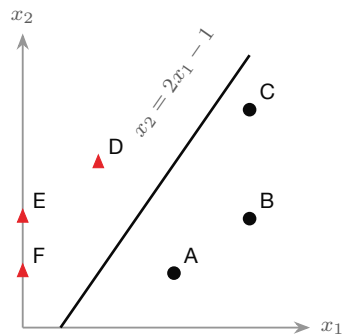
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The results

pt	(x_1, x_2)	score	predicted	label
A	$(2, 1)$	$+2$	$+1$	$+1$
B	$(3, 2)$	$+3$	$+1$	$+1$

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6 of 6 correct — boundary $x_2 = 2x_1 - 1$.

One neuron = one line.

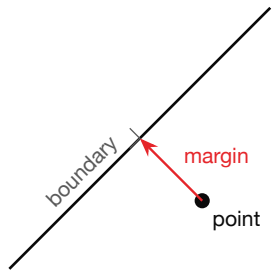
Now — how **good** is that line?

How confident? The margin

Margin

$$\text{margin} = \frac{\mathbf{w} \cdot \mathbf{x} + b}{\|\mathbf{w}\|}$$

$\|\mathbf{w}\| = \sqrt{w_1^2 + w_2^2}$ — the length of the weight arrow.

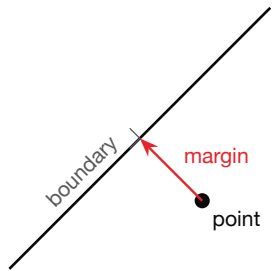


How confident? The margin

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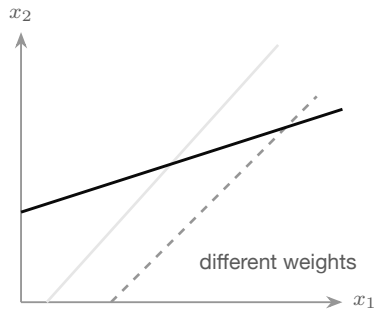
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A: margin $2/\sqrt{5} \approx 0.89$ C: margin $1/\sqrt{5} \approx 0.45$

Change the weights, watch the line move



Grading the classifier

Precision

$$\frac{\text{TP}}{\text{TP} + \text{FP}}$$

Recall

$$\frac{\text{TP}}{\text{TP} + \text{FN}}$$

TP = true positive, FP = false positive, FN = false negative; positive = the class we approve

Grading the classifier

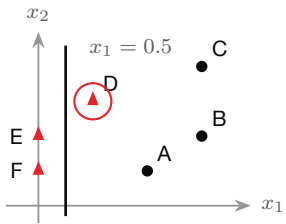
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a different line: $\mathbf{w} = [1, 0]$, $b = -0.5$

TP $= 3$ (A, B, C) FP $= 1$ (D) FN $= 0$

Grading the classifier

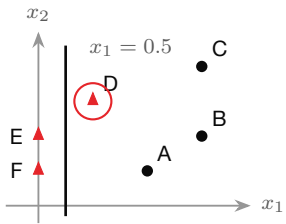
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precision $= \frac{3}{4} = 75\%$ recall $= \frac{3}{3} = 100\%$

Every true case caught. One wrong approval.

A neuron as maths

The weighted sum

Two classifiers, one dataset

The pattern it could not learn

One machine, two limits

Two classifiers, one dataset

same data, two very different machines

Two classifiers, one dataset

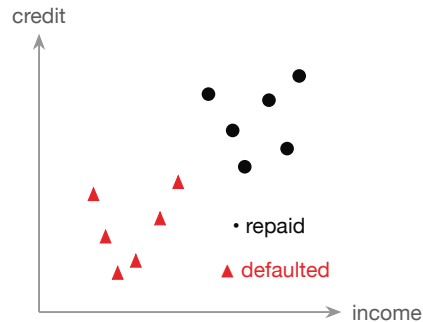
Loan applications. Two features:

- an **income score**
- a **credit score**

Label: **repaid** $(+1)$ or **defaulted** (-1).

350 applications to train on, **150** set aside for testing.

Naive Bayes: binned low / medium / high + Laplace | weighted sum: raw scores.



Poll

Two equally-accurate models make opposite mistakes: one over-rejects safe applicants, the other over-approves risky ones.

Who is left worse off when a safe applicant is wrongly rejected?

- **The bank that loses a good customer**
- **The applicant refused a loan they could repay**
- **Future applicants, if the rejection trains the next model**
- **All of these, but not equally**

Vote now — scan the code for your campus, *or* type the address.



Melbourne

pollev.com/juliangarcia



Malaysia

pollev.com/onghueyfang767

Same data, opposite mistakes

	accuracy	precision	recall	false approvals	false rejections
Naive Bayes	81%	77%	**96%**	**25**	3

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Same 150-row test set for both: 85 repaid, 65 defaulted.

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Similar accuracy, opposite errors

Naive Bayes: **permissive**. Weighted sum: **conservative**.

A neuron as maths

The weighted sum

Two classifiers, one dataset

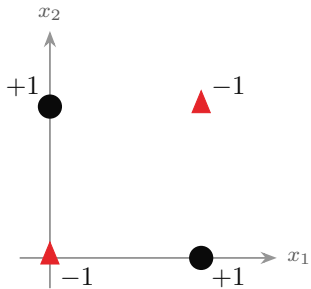
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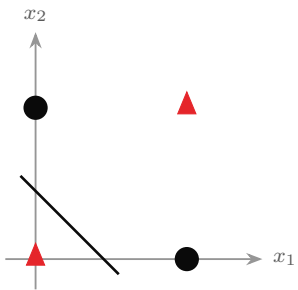
one straight line is not always enough

Four points a line cannot split

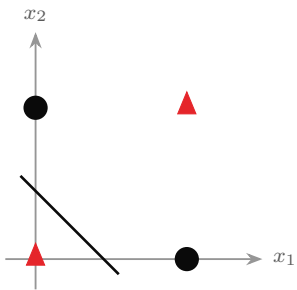


Can one **straight line** put both $+1$s on one side and both -1s on the other?

Try a line — any line

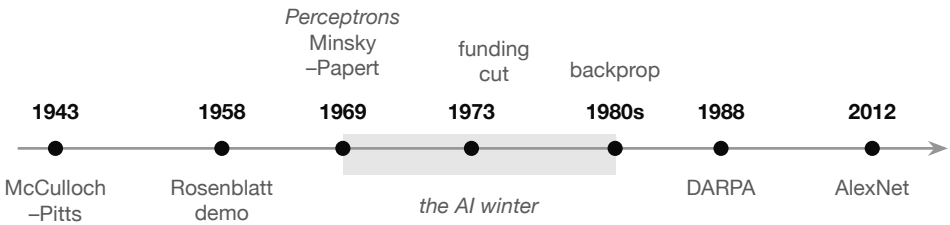


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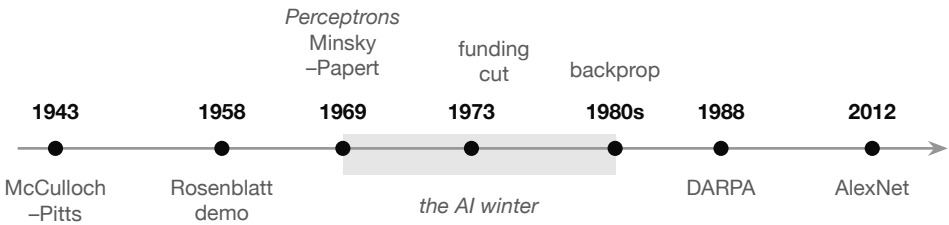


The best any single line manages is **3 out of 4**.

The proof — and the collapse



The proof — and the collapse



Minsky & Papert, *Perceptrons* (1969) — on stacking layers: **“sterile.”** 1973: UK funding **ends**; US support falls away too.

The revival — in three stages

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Key Takeaway

Every modern network — image classifiers, language models — is perceptrons **stacked in layers**.

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One machine, two limits

a line it cannot draw — and a group it can harm

Poll

Two groups have very different rates of repaying loans. A single classifier is “accurate overall”.

Can it still work worse for one group?

- **Yes**
- **No**
- **Only if the data is wrong**

Vote now — scan the code for your campus, *or* type the address.



Melbourne

pollev.com/juliangarcia



Malaysia

pollev.com/onghueyfang767

Accurate — for whom?

	base repay rate
Group A	~79%
Group B	~38%

Two demographic groups of applicants in the loan data. These are the rates the data already shows, before any model touches it.

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Discussion

When the base rates differ this much, “accurate overall” can still mean **worse for one group**.

Who should bear each kind of mistake — and who decides?

In the lab this week

Pass tier. **Aiming for a Pass**

1. Build the weighted sum + threshold, and predict ± 1.
2. Find the decision boundary; measure a prediction's confidence (the margin).
3. Check precision & recall.
4. See where a single straight line fails (the opposite-corners pattern, XOR).

Credit tier. **Aiming for a Credit** — all of the above, plus:

- Put **two classifiers** on the same data and compare *which mistakes* each one makes.

Distinction tier. **Aiming for a Distinction** — all of the above, plus:

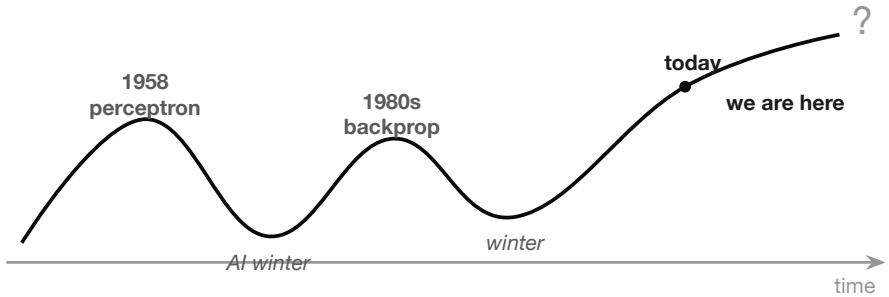
- Think about how a machine could **tune its own weights** from its mistakes — the step this week deliberately leaves open.

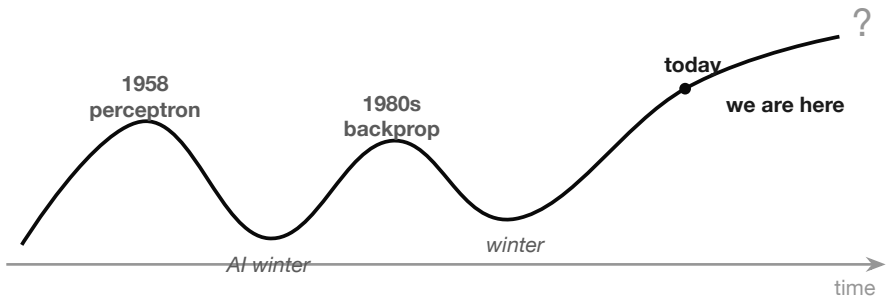
You built the **deciding**.

You built the **deciding**.

Next: **who the decisions affect**.

Then the **learning** — the machine tunes its own weights.


?
today
we are here
1958
perceptron
1980s
backprop
AI winter
winter
time



Boom, bust, boom — the pattern repeats.